DR. ISHRAT–UL–EBAD KHAN INSTITUTE OF ORAL HEALTH SCIENCES
DOW UNIVERSITY OF HEALTH SCIENCES
DEPARMENT OF ANATOMY-1ST YEAR BDS–BCQ CLASS TEST

Name: _____

Roll no: _____

Max Marks: 28
Min Marks: 14

Date: 09-03-2024

Each BCQ carries one (01) mark. There will be no negative marking.

Q1) Which of the following is an acidic stain?

a) Basic fuchsin
b) Eosin
c) Hematoxylin
d) Methylene blue
e) Toluidine blue

Q2) Which of the following statements regarding cell membrane is false?

a) It has a trilaminar structure.
b) It is principally made up of phospholipids, proteins and cholesterol.
c) Its glycocalyx is formed by glycolipids and glycoproteins.
d) Its thickness ranges from 7.5 to 10 nm.
e) Transmembrane protein is an example of peripheral protein.

Q3) Mark the correct statement about different organelles of the cell:

a) Cristae are the infoldings of the outer membrane of mitochondria.
b) Centrioles are made up of microtubules.
c) Endoplasmic reticulum is studded with lysosomes on its surface.
d) Lysosomes are bounded by double membrane.
e) Nuclear envelope is connected to the golgi apparatus.

Q4) Which of the following is not a membrane-bound organelle?

a) Golgi apparatus
b) Mitochondrion
c) Ribosome
d) Rough endoplasmic reticulum
e) Smooth endoplasmic reticulum

Q5) Which of the following is not a feature of the primary lysosomes?

a) They are bounded by a single membrane.
b) They carry out intracellular digestion.

c) They contain acid hydrolases.
d) They contain the DNA.
e) They emerge from the golgi apparatus.

Q6) A nucleus may appear vesicular, most likely due to the presence of large amount of the:

a) Euchromatin
b) Heterochromatin
c) Lipids
d) Lysosomes
e) Ribosomes

Q7) Sequestration, packaging, and distribution of proteins is mainly carried out by which cell organelle?

a) Golgi apparatus
b) Mitochondria
c) Ribosome
d) Rough endoplasmic reticulum
e) Smooth endoplasmic reticulum

Q8) When a junctional complex is spot-like and there is an intercellular gap of about 30 nm between the neighboring cells linked together by transmembrane proteins, it is called the:

a) Desmosome
b) Fascia adherens
c) Hemidesmosome
d) Zonula adherens
e) Zonula occludens

Q9) The adhering junction which binds the epithelial cells to their underlying basal lamina is called the:

a) Desmosome
b) Hemidesmosome
c) Nexus junction
d) Zonula adherens
e) Zonula occludens

Q7) Alveoli of the lung are lined by the:

a) Pseudostratified columnar epithelium
b) Simple cuboidal epithelium
c) Simple squamous epithelium
d) Stratified columnar epithelium
e) Stratified cuboidal epithelium

Q8) Stratified squamous keratinized epithelium is found in the:

a) Conjunctiva of the eye
b) Esophagus
c) Gall bladder
d) Epidermis of skin
e) Trachea

Q9) Urinary bladder has a lining of the:

a) Pseudostratified columnar epithelium
b) Simple columnar epithelium
c) Stratified squamous keratinized epithelium
d) Stratified squamous non-keratinized epithelium
e) Transitional epithelium

Q10) 'Striated border' appearance of small intestinal mucosal cells on microscopic examination is due to the presence of which cell surface modification?

a) Cilia
b) Flagella
c) Microvilli
d) Stereocilia
e) Villi

Q11) The internal skeleton of a microvillus is made up of a longitudinal central bundle of the:

a) Actin filaments
b) Desmin filaments
c) Keratin filaments
d) Microtubules
e) Myosin filaments

Q12) The axoneme of a cilium mainly comprises of the:

a) Actin filaments
b) Desmin filaments
c) Keratin filaments
d) Microtubules
e) Myosin filaments

Q13) Which of the following is an example of branched simple tubular gland?

a) Intestinal glands
b) Principal glands of stomach
c) Sebaceous glands
d) Sweat glands



e) Submandibular glands

Q14) Which of the following different glands follow holocrine mode of secretion?

a) Intestinal glands
b) Lacrimal glands
c) Salivary glands
d) Sebaceous glands
e) Sweat glands of axilla

Q15) A plane passing through the center of the body dividing it into equal right and left halves is called:

a) Coronal plane
b) Median plane
c) Sagittal plane
d) Oblique plane
e) Transverse pane

Q16) A rescue team approaches an unconscious victim lying on his chest and abdomen with his back facing upwards, on the accident site. The unconscious person is said to be in the:

a) Anatomical position
b) Fundamental position
c) Lithotomy position
d) Prone position
e) Supine position

Q17) When the angle between two articulating bones increases, it is termed as:

a) Abduction
b) Adduction
c) Extension
d) Flexion
e) Medial rotation

Q18) Regarding different terms of relation of body parts and their meanings, mark the incorrect pair:

a) Contralateral----------on the same side of body.
b) Cranial----------towards/close to the head.
c) Distal----------away from the root.
d) Evagination----------projection outside.
e) Superficial----------near the skin/surface of the body.

Q19) Which of the following is an example of a modified long bone?

a) Clavicle

b) Femur
c) Maxilla
d) Mandible
e) Scapula

Q20) Which of the following is a salient feature of the pneumatic bones?

a) They appear as small bony nodules.
b) They contain air-filled spaces.
c) They are not lined by the periosteum.
d) They are found under the tendons.
e) They minimize the friction.

Q21) In which part of a typical long bone the length-wise growth occurs?

a) Diaphysis
b) Epiphyseal plate of cartilage
c) Epiphysis
d) Metaphysis
e) Nutrient foramen

Q22) The periosteal arteries supply which area/part of a long bone?

a) Epiphysis
b) Inner 2/3rd of cortex
c) Medullary cavity
d) Metaphysis
e) Outer 1/3rd of cortex

Q23) Separation of the sister chromatids and their movement to their respective poles occurs in which stage of mitosis?

a) Anaphase
b) Cytokinesis
c) Metaphase
d) Prophase
e) Telophase

Q24) The phenomenon of crossing-over (exchange of genetic material) takes place in which sub-stage of prophase of meiosis-1?

a) Diakinesis
b) Diplotene
c) Leptotene
d) Pachytene
e) Zygotene

Q25) In which stage of prophase-1 does the primary oocyte remain arrested till puberty?

a) Diakinesis
b) Diplotene
c) Leptotene
d) Pachytene
e) Zygotene

Q26) Primary oocyte with zona pellucida surrounded only by simple or stratified layer of follicular/epithelial cells is appropriately termed as:

a) Antral follicle
b) Graafian follicle
c) Oogonium
d) Primary follicle
e) Primordial follicle

Q27) Regarding spermatogenesis, mark the odd/false statement:

a) Primary spermatocytes contain haploid number of chromosomes.
b) The developing spermatozoa are supported by sertoli cells.
c) Type-A pale spermatogonia give rise to type-B spermatogonia.
d) Type-B spermatogonia differentiate into primary spermatocytes.
e) Time required for spermatogenesis to complete is about 74 days.

Q28) The morphological changes that occur during spermiogenesis do not include:

a) Completion of meiosis-1.
b) Condensation of nucleus.
c) Formation of acrosome.
d) Formation of neck, middle piece and tail.
e) Shedding of most of the cytoplasm.

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